International Games Security Summit

Skipping the Security Side Quests: A Qualitative Study on Security Practices and Challenges in Game Development

Philip K., **Sabrina K.**, Sandra H., Alexander K., Niklas B., Sascha F.
CISPA, University of Hannover

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Hornella SOUDE
Tristan LECONTE--DENIS

Summary

1 Motivation & Research Approach

2 Key Findings from Industry Interviews

3 Practical Implications & Takeaways

Motivation & Research Approach

Why this matters

In recent years, the video game industry had an exponential growth in popularity.

- Games are **huge businesses** and **attractive attack targets**.
- Modern “Games-as-a-Service” add online features: **auth**, **payments**, **stores**.
- Threats: **cheating**, **fraud**, data leaks, **RCE**, downtime**.**

→ How do big video game studios **include security** in their **development process**?

Figure 1 – Flaticons Game Controller

Research questions & method

The method includes **20 semi-structured interviews**, mixed roles (**devs**, **producers**, QA, **security**)

Awareness & Priority

- How aware are developers of different security areas?
- How do they prioritize them?

Studios challenges

- What are the main challenges that studios face that prevent secure development?

Methods & Guidelines

- What tools, processes or guidelines are currently used in practice by game studios to include security?

Big pictures of findings

Game developers are aware
Video games developers are aware that security matters in their workflow but faces challenges:

- **Time pressure**
  Video game development have strict deadlines
- **Economic constraints**
  The budget often doesn't include security measures
- **Nature of the game industry**
  Video game industry is hit-driven

Figure 2 – Flaticons Challenge

Key Findings for Industry Interviews

Five security areas – 1/3

Anti-cheat

Protecting competitive integrity and revenue

Asset security

Safeguarding intellectual property (code, art, audio)

Network & backend

Ensuring service availability and secure transactions

Game client security

Preventing technical exploits like RCE or injections

User privacy & telemetry

Managing player data and legal compliance

Five security areas – 2/3

Anti-cheat

Daily battle

Cheating affects player trust and revenues.

Third-party solutions

Most developers relies on 3rd party software to detect cheating.

> **Consume significant development resources.**

Asset security

Intellectual property

Developers worry about stolen code, assets or cloned games.

Solutions

Code obfuscation or encrypted assets.

> **Full protection is unrealistic?**

Network & backend

Third-party SDKs

Modern games rely on SDKs (auth, payments, analytics, etc) without auditing the library.

Encryption

Encryption of communication but with a priority toward performance.

> **Performance over security**

Five security areas – 3/3

Game client

Stability

Stability is prioritized over code security which leads programming flaws within the client

Vulnerability

Most developers use engines and custom code together leading to vulnerabilities

> **Full protection is unrealistic?**

Privacy & telemetry

Data collections

Most studios collects gameplay telemetry (e.g., to improve balancing, monetization, etc..) which involve more external libraries

GPDR

Mentioned as an external driver to improve security/privacy but achieved through external tools rather that internal expertise

Why it happens

The main issue across all five areas is the structural challenges of the game industry.

- **Hit-driven industry** (rapid prototypes, short lifecycle, feature pressure)
- **Disconnect** between **publishers**, **managers**, **engineers** on technical security
- **Poor guidance** and **training** on security
- Tight deadlines, limited budget, small teams, etc.

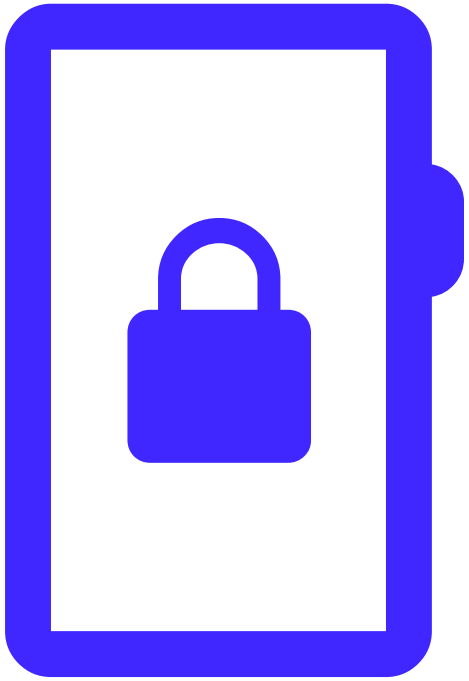


Figure 3 – Google icon about security

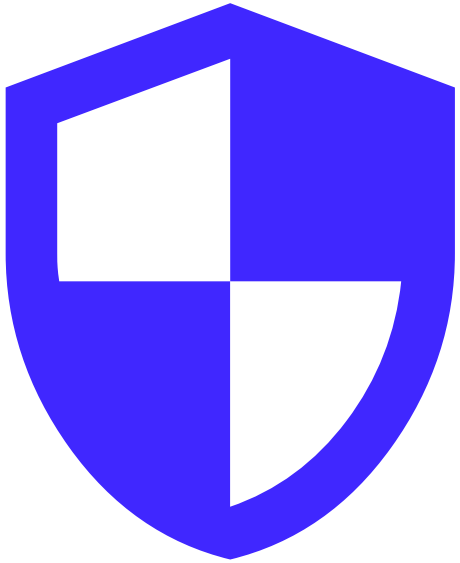
As a summary, **tight deadlines**, **limited budget**, **small teams** and **lack of training** leads to a lack of security.

Practical Implications & Takeaways

Recommendations

Low-efforts / high-impact

- Adopt server-authoritative checks for sensitive actions
- User dependency analysis to catch know library vulnerabilities
- Prefer well-maintained third-party services
- Simple policies: minimal telemetry, data minimization, documented incident paths, etc.
- Invest in brief security training for developers



Conclusion

"Game security proved to be a complex landscape under intense pressure and heavy resource constraints."

→ Studios want **secure games**, but **economics** and **pace push** security to the **side quest**. The paper argues for more knowledge transfer between **security researchers** and the **games industry,** so these steps become standard practice.

Thank you!